

Object Detection in Computer Vision

Abstract

Object detection is a fundamental computer vision task that involves identifying and localizing objects within images. This paper reviews modern deep learning approaches including YOLO, Faster R-CNN, and transformer-based detectors, comparing accuracy and inference speed.

Region-Based Methods

Region-based convolutional neural networks (R-CNN) introduced a two-stage pipeline: first generate region proposals using selective search, then classify each region using a CNN. Faster R-CNN improved this by introducing the Region Proposal Network (RPN).

Single-Stage Detectors

YOLO (You Only Look Once) reformulates object detection as a single regression problem, dividing the image into a grid and predicting bounding boxes and class probabilities directly. This enables real-time detection at 45 frames per second.

Evaluation Metrics

Detection performance is measured using mean Average Precision (mAP) across different IoU thresholds. The COCO benchmark uses mAP averaged over IoU thresholds from 0.5 to 0.95, rewarding precise localization.